화학특강 (2012년 2학기)
제1장 화학의 기본 개념

중요 학습 내용

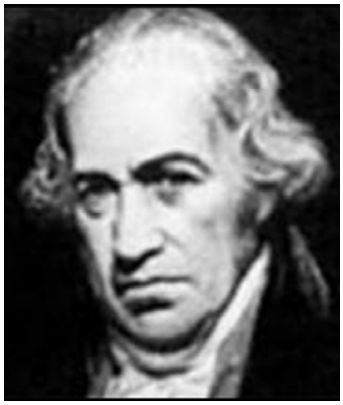
- 물의 구조와 특성
- 수소결합
- 용액의 성질: 확산계수와 점성도
- 화학퍼텐셜

Units

1. Units
   1) SI prefix (metric prefix) (SI = Systemè International)

SI prefix	Symbol	Power of 10	SI prefix	Symbol	Power of 10
peta	P	10^{15}	milli	m	10^{-3}
tera	T	10^{12}	micro	μ	10^{-6}
giga	G	10^{9}	nano	n	10^{-9}
mega	M	10^{6}	pico	p	10^{-12}
kilo	k	10^{3}	femto	f	10^{-15}
centi	c	10^{-2}	atto	a	10^{-18}

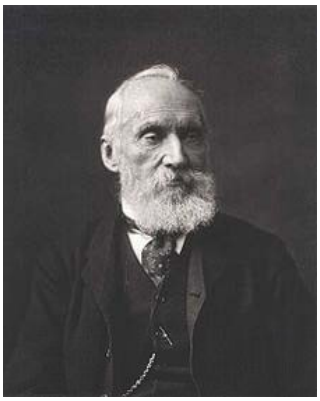
섭씨 온도, 화씨 온도, 그리고 절대 온도



Daniel Gabriel Fahrenheit,
華倫海 (화륜해)
(1686–1736),
독일의 물리학자, 공학자



Anders Celcius,
攝爾思 (섭이사)
(1701–1744),
스웨덴 천문학자



William Thomson
"Lord Kelvin"
(1824 - 1907),
영국의 물리학자, 공학자

	섭씨 온도 (°C)	화씨 온도 (°F)	절대 온도 (K)
물의 어는점	0	32	273.15
물의 끓는점	100	212	373.15

$$^{\circ}\mathrm{C} = \frac{5}{9}(^{\circ}\mathrm{F} - 32)$$

$$\mathrm{K} = ^{\circ}\ \mathrm{C} + 273.15$$

Units

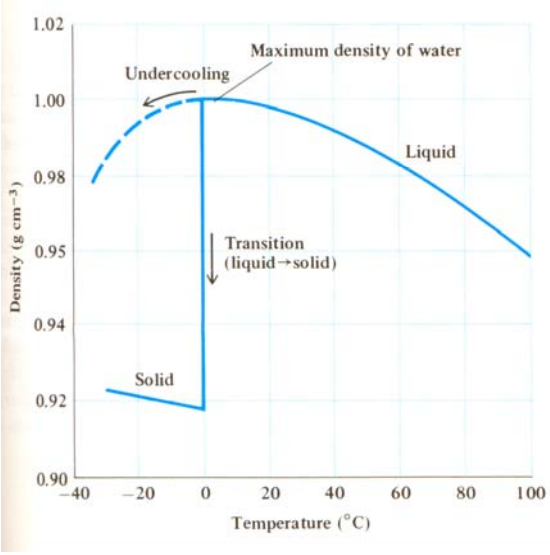
1. Units
   1) Volume
      - 1 mL = 1 cm^3 = 1 cc
      - 1 gallon = 3.785 L = 8 pint
   2) Mass and weight
      - 1 lb ≈ 0.453 kg
   3) Density
      - Density = Mass / Volume
      - Water density = 0.99707 g/cm^3 (25°C)



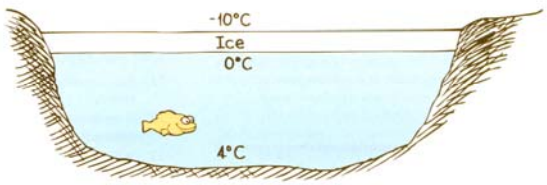
(예제 1) 25°C에서 물의 밀도를 1.00 g/cm^{3}라고 할 때, 물의 몰농도를 구하여라.

Water density

- Water density



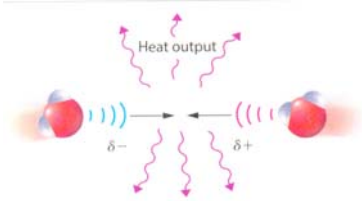
- Why do lakes freeze from the top down?



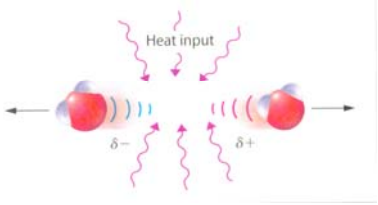
1 기압 하에서 얼음의 융해 과정에 대한 엔탈피, 엔트로피, 그리고 Gibbs 자유에너지 변화

T (K)	ΔH_{fus} (kJ mol^{-1})	ΔS_{fus} (kJ mol^{-1}K^{-1})	ΔG_{fus} (kJ mol^{-1})
263	5.619	0.0206	0.213
273	6.008	0.0220	0.0
283	6.397	0.0234	-0.226

물의 융해 과정에서 ΔH와 ΔS가 0보다 큰 이유는?

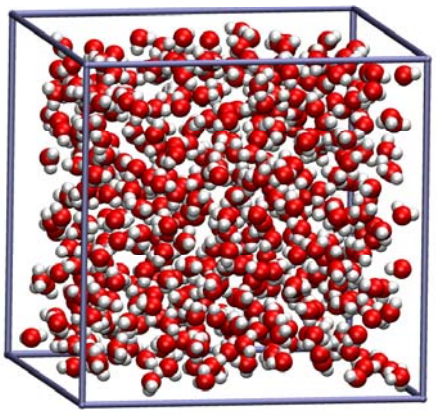


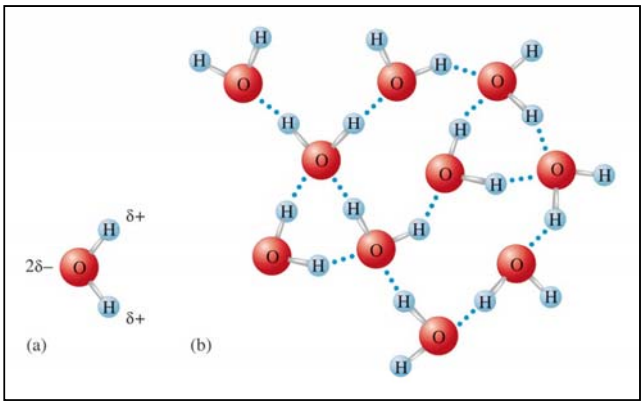
얼음이 되는 경우 열 방출 ($\Delta H < 0$)



물이 되는 경우 열 흡수 ($\Delta H > 0$)

Structure of Liquid Water

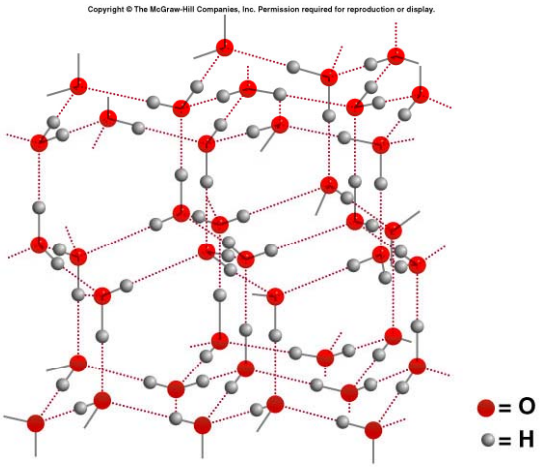




액체 상태에서 물분자들의 배열이 얼음 상태에서 물분자들의 배열보다 더 무질서하므로 액체 상태 물의 엔트로피가 더 크다. 점선은 산소원자와 수소원자 간의 수소결합을 나타낸다.

Structure of Iced Water (I)

- Three-dimensional structure of ice

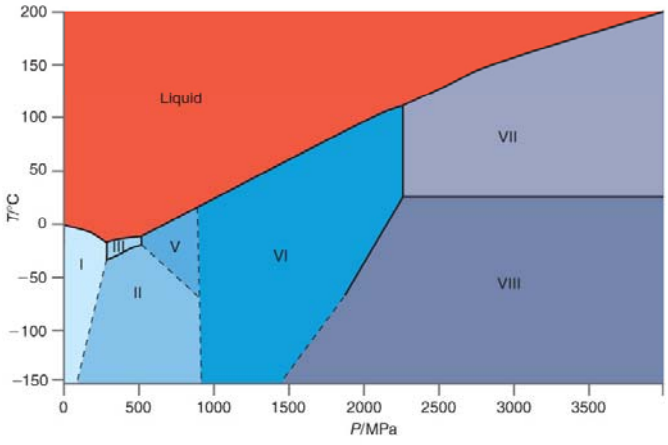


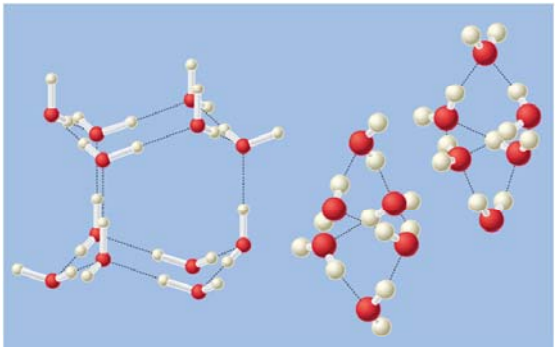


얼음의 3 차원 입체 구조. 한 개의 산소 원자는 네 개의 수소 원자와 공유 결합 혹은 수소 결합에 의해 연결되어 있다. 점선은 산소 원자와 수소 원자 간의 수소 결합을 나타낸다.

Structure of Iced Water (II)

- 온도와 압력에 따른 물의 서로 다른 고체상: 얼음의 여러가지 구조

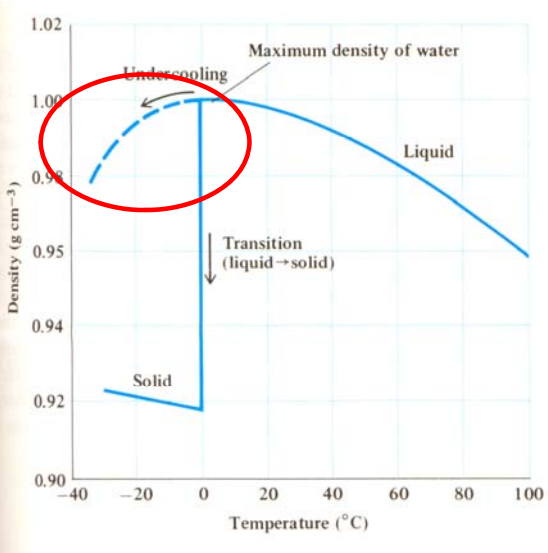




물의 서로 다른 두 가지 결정구조: 얼음 I과 얼음 VI

육방 얼음 (얼음 I)의 밀도는 0°C, 1기압에서 0.9167 g/cm^{3}이고 얼음 VI의 밀도는 1.6 GPa에서 1.31 g/cm^{3}이다. 이 압력에서 액체물의 밀도는 1.18 g/cm^{3}이다. 얼음 VI이 물에 뜰까? 가라앉을까?

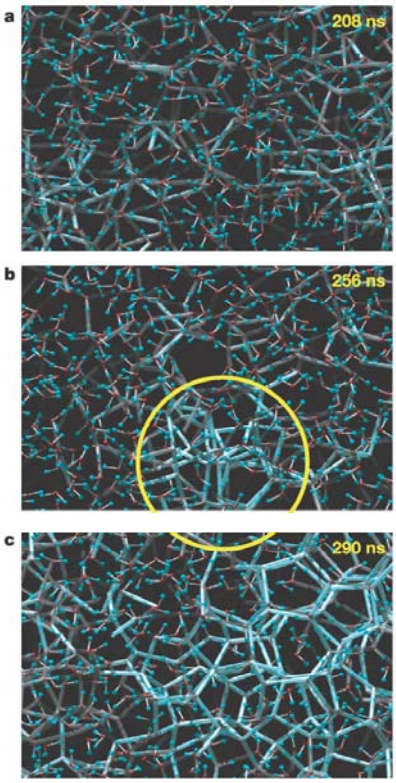
Supercooled Water

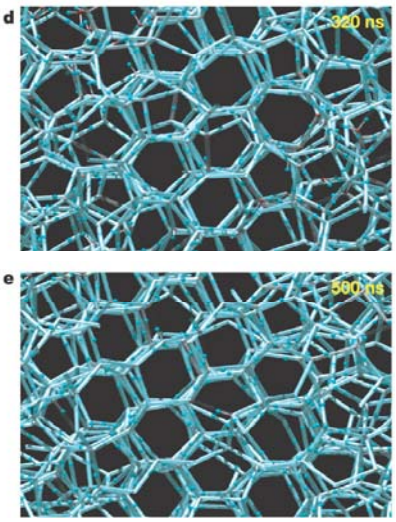






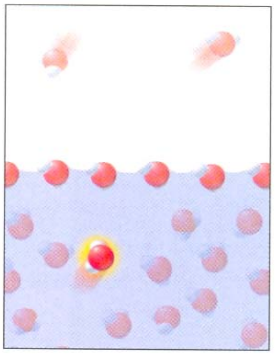
Supercooled water: Molecular dynamics simulation (분자동역학 모의실험)



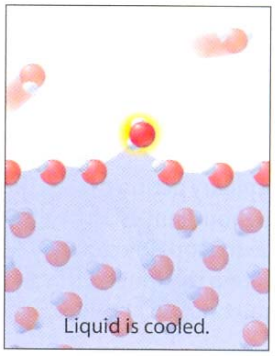


Matsumoto et al., Nature **416**, 409 (2002)

Water vaporization



수소 결합을 끊을 수 있을 정도의 운동 에너지를 가진 물 분자가 표면으로 이동



높은 운동 에너지를 가진 물 분자가 공기 중으로 날아갈 때 물의 온도는 낮아진다.



운동 에너지를 잃은 기체 상태의 물 분자를 받아들인 공기의 온도도 낮아진다.





Normal boiling points and standard enthalpy changes and standard entropy changes of vaporization

Table 18.1 Normal boiling points and standard enthalpy changes and standard entropy changes of vaporization

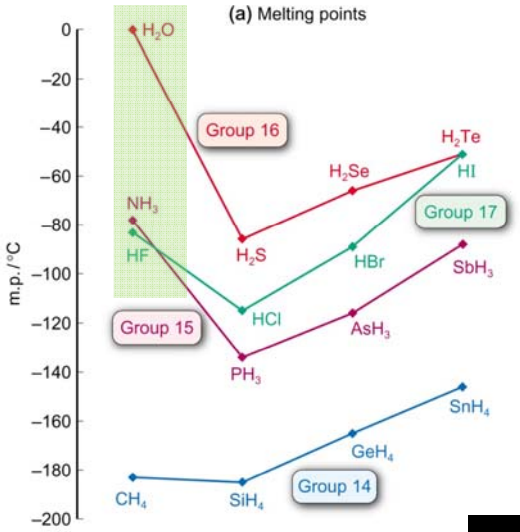
	b.p./°C	$\Delta_{vap}H^{\ominus}$/kJ mol^{-1}	$\Delta_{vap}S^{\ominus}$/J K^{-1} mol^{-1}
Benzene (C_6H_6)	80.1	+30.8	+87.2
Carbon disulfide (CS_2)	46.3	+26.7	+83.7
Hydrogen sulfide (H_2S)	−60.4	+18.7	+87.9
Tetrachloromethane (CCl_4)	76.7	+30.0	+85.8
Cyclohexane (C_6H_{12})	80.7	+30.1	+85.1
Decane ($C_{10}H_{22}$)	174	+38.8	+86.7
Methoxymethane ($(CH_3)_2O$)	−23.0	+21.5	+86.0
Methanol (CH_3OH)	65.0	+35.2	+104.1
Ethanol (C_2H_5OH)	78.3	+38.6	+110.0
Water (H_2O)	100.0	+40.7	+109.1
Butane (C_4H_{10})	−0.50	+22.4	+82.1
Methylbenzene ($C_6H_5CH_3$)	110.7	+33.5	+87.2
Ethanoic acid (CH_3COOH)	118.3	+24.4	+61.9

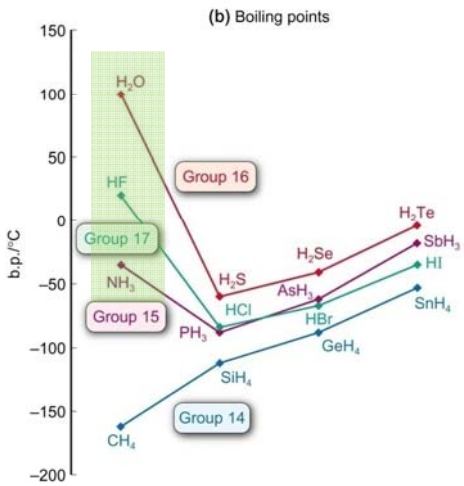
Trouton's rule:
$\Delta S_{vap}^{\circ} \approx +85$ J K^{-1} mol^{-1}

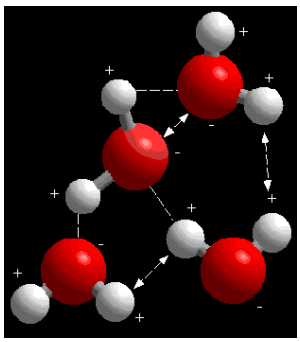
Periodic Table

Group → / ↓ Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Uut	114 Uuq	115 Uup	116 Uuh	117 Uus	118 Uuo
란탄넘 계열				57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
악티늄 계열				89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr

The Hydrogen Bond







Origin of the Hydrogen Bond

1. Large electronegativity difference between H and F, O, and N.
   - Electronegativity: the ability of an atom to attract toward itself the electrons in a chemical bond.
   - Linus Pauling

$$\chi_A - \chi_B = (eV)^{-1/2}\sqrt{E_d(AB) - [E_d(AA) + E_d(BB)]/2}$$

Electronegativity of HBr = 0.73
(dissociation energies:
H–Br, 3.79 eV; H–H, 4.52 eV; Br–Br 2.00 eV)

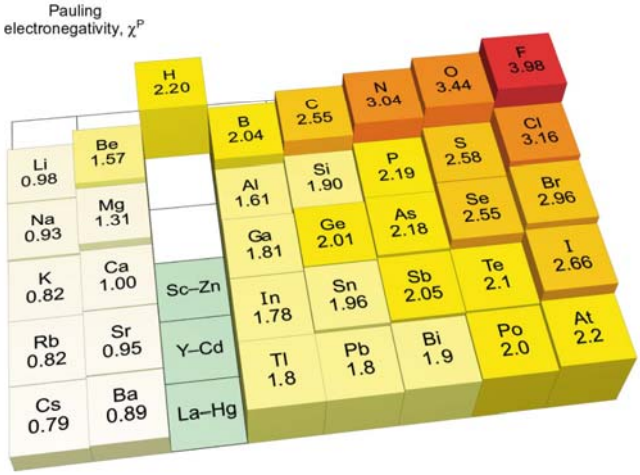


TABLE 2.2 Bond Dissociation Energies of Diatomic Molecules ($kJ \cdot mol^{-1}$)

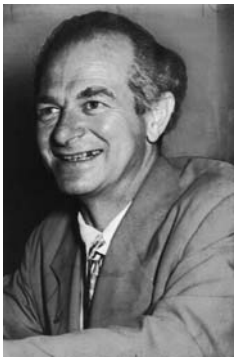
Molecule	Bond dissociation energy
H_2	424
N_2	932
O_2	484
CO	1062
F_2	146
Cl_2	230
Br_2	181
I_2	139
HF	543
HCl	419
HBr	354
HI	287

1 eV = 96.485 kJ mol^{-1}

Chang, 8.3, 9.5, 9.10, and 11.2



Marie Skłodowska Curie
Physics in 1903
Chemistry in 1911



Linus Pauling
Chemisty in 1954
Peac Prize in 1962



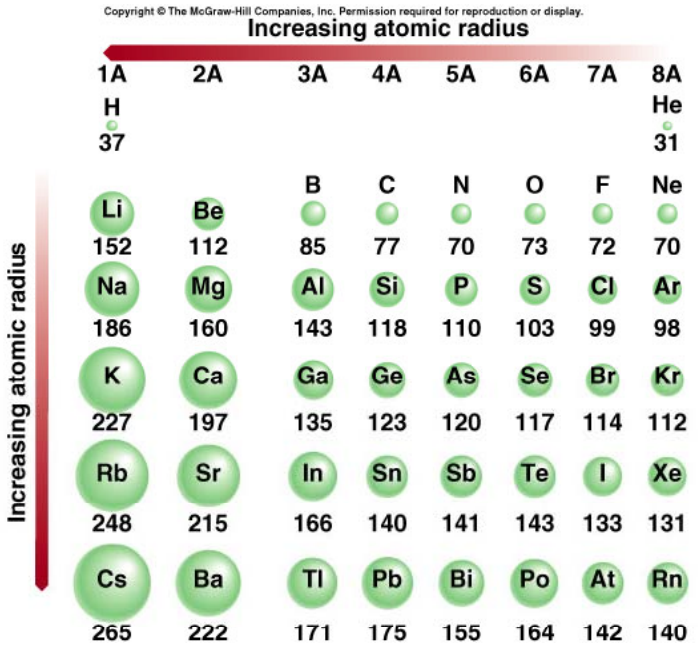
John Bardeen
Physics in 1956
Physics in 1972



Frederic Sanger
Chemistry in 1958
Chemistry in 1980

Origin of the Hydrogen Bond

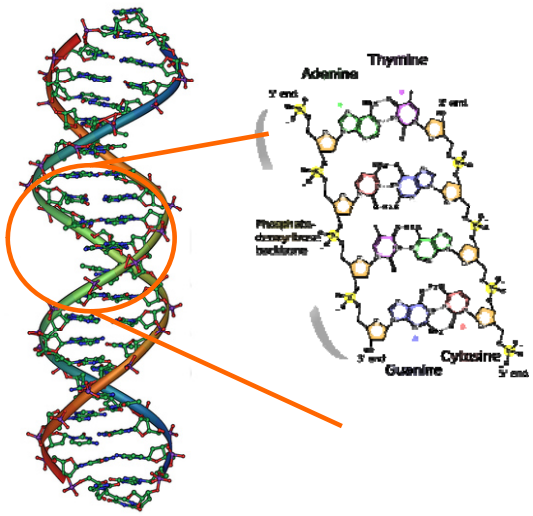
2. Small size of F, O, and N.



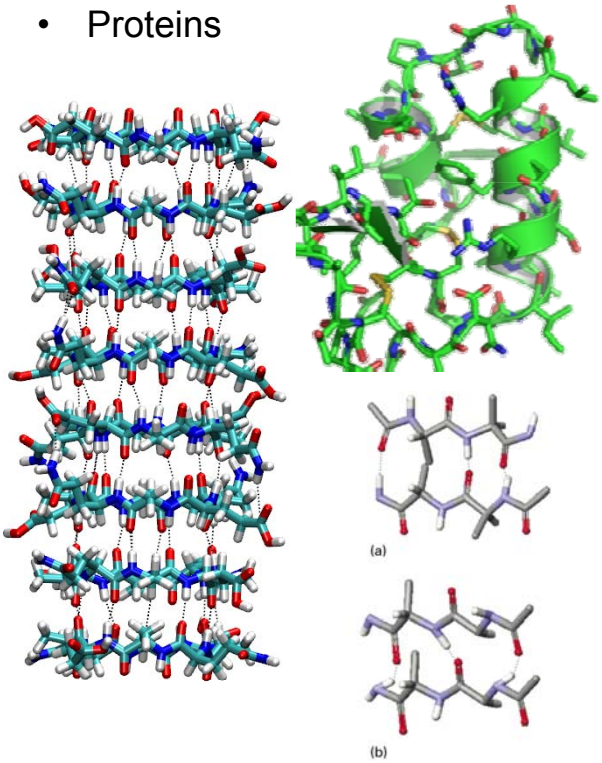
Chang, 8.3, 9.5, 9.10, and 11.2

Hydrogen bonds in biological systems

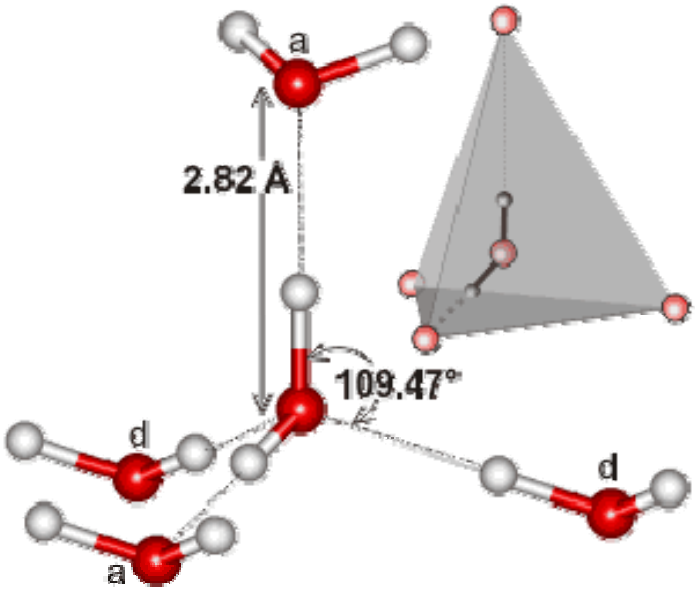
- DNA double helix



- Proteins



Structure of hydrogen bonds in water



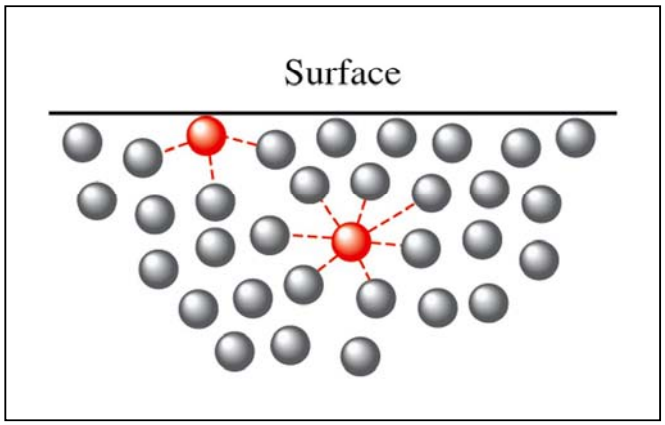
Water Hydrogen Bond and Surface Tension

- 표면장력: 액체의 표면을 단위면적만큼 증가시키는데 필요한 자유에너지.

$$\gamma = \frac{dA}{d\sigma}$$







액체 표면에 있는 분자들은 내부에 있는 분자들보다 끌어당김 상호작용 에너지가 더 작고 그 결과 더 불안정해진다.

Chang, 11.3

Surface tension of liquids, Laplace equation, and Kelvin equation

Table 17.5* Surface tensions of liquids at 293 K

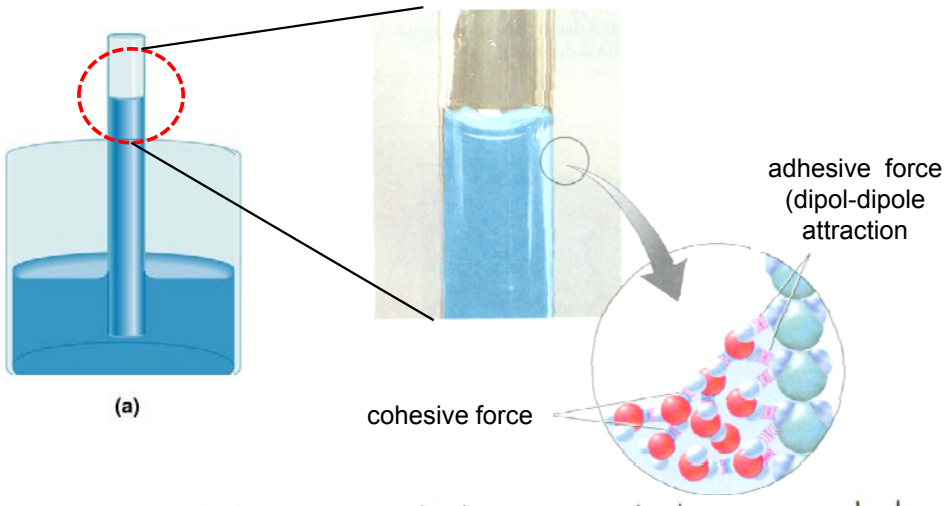
	$\gamma/(\text{mN m}^{-1})$
Benzene	28.88
Mercury	472
Methanol	22.6
Water	72.75

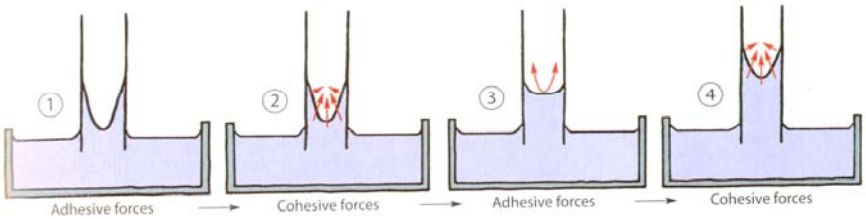
* More values are given in the *Data section*.
Note that $1\ \text{N m}^{-1} = 1\ \text{J m}^{-2}$.

Atkins, 9th Ed. p646

Cohesion and adhesion

- cohesion and adhesion
  - cohesion: the intermolecular attraction between like molecules.
  - adhesion: an attraction between unlike molecules.





Chang, 11.3

화학 퍼텐셜 (chemical potential)

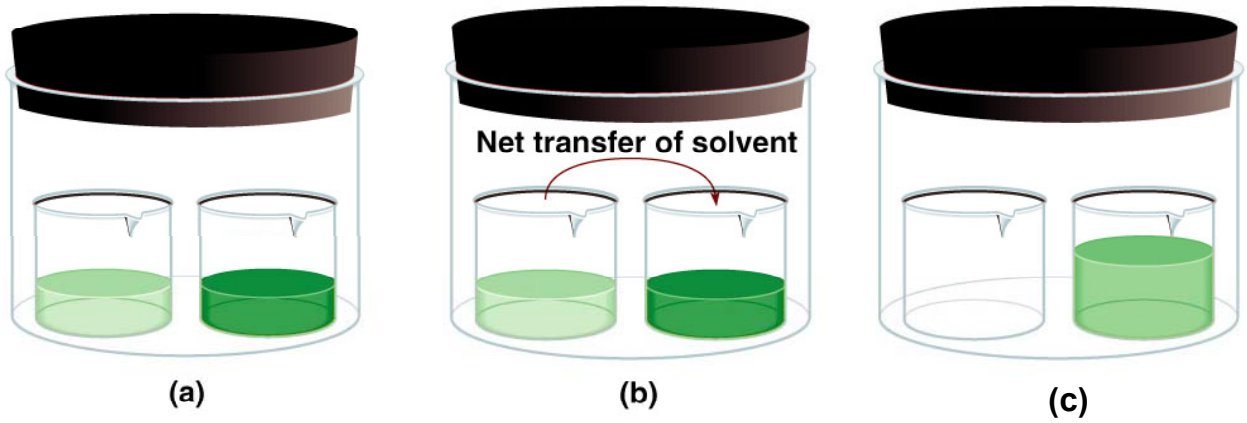
- 화학 퍼텐셜: 일정한 상에 있는 물질이 다른 상으로 움직이려는 경향.
- 물질은 화학 퍼텐셜이 높은 상에서 낮은 상으로 움직이려는 경향이 있다.
- 이상기체의 화학 퍼텐셜

$$\mu(T,P) = \mu^{\circ}(T) + RT\ln\left(\frac{P}{P^{\circ}}\right)$$

- 이상액체의 화학 퍼텐셜

$$\mu_{\text{용매}}(T,P) = \mu^{*}_{\text{용매}}(T,P) + RT\ln x_{\text{용매}}$$

화학퍼텐셜



Chang, 12.6

Laplace equation and Kelvin equation for curved surfaces

- Laplace equation

$$P_{\text{in}} = P_{\text{out}} + \frac{2\gamma}{r}$$

- Kelvin equation

$$P = P^* \exp\left[\frac{2\gamma \bar{V}(l)}{rRT}\right]$$

Equilibrium Vapor Pressure Increase Over Pure Water Droplet as a Function of Droplet Radius at T = 25 C						
r_p (μm)	1	0.3	0.1	0.03	0.01	0.003
P/P_0	1.0011	1.0035	1.0107	1.0360	1.1118	1.4238
ΔP (%)	+0.11	+0.35	+1.1	+3.6	+11	+42

Equilibrium Vapor Pressure Decrease of Pure Water inside a Pore as a Function of Pore Radius at T = 25 C						
r_p (μm)	1	0.3	0.1	0.03	0.01	0.003
P/P_0	0.9989	0.9964	0.9895	0.9653	0.8994	0.7023
ΔP (%)	−0.11	−0.35	−1.1	−3.5	−10	−30

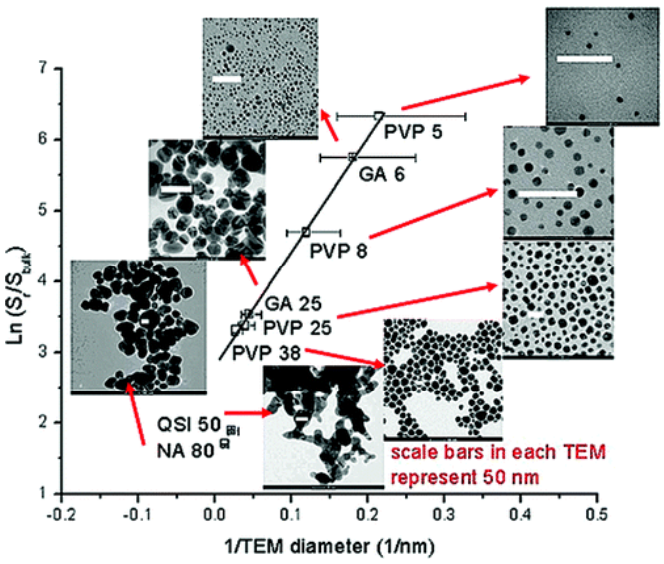
Solubility of ions and Kelvin equation

- Kelvin equation for solubility

$$x_A = x_A^* \exp\left[\frac{2\gamma \bar{V}(s)}{rRT}\right]$$

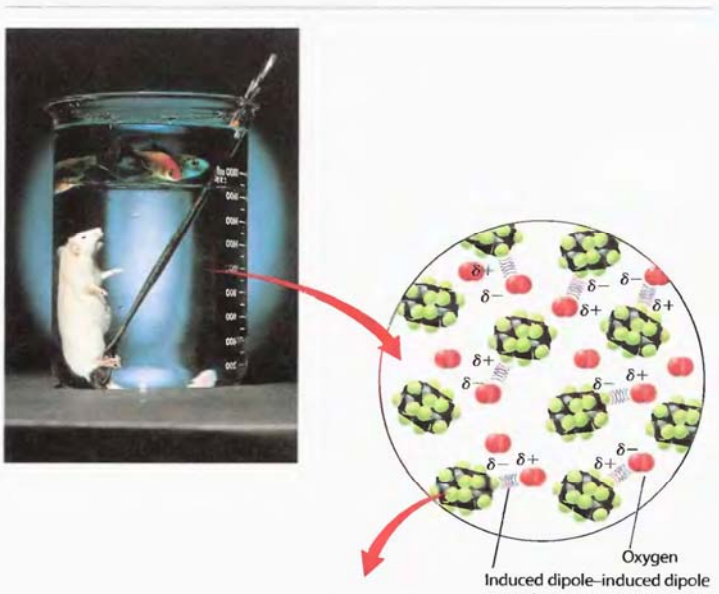
x_A^*: mole fraction of A with a flat surface
x_A: mole fraction of A with radius r

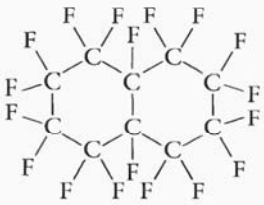
- solubility of silver nanoparticles



Ma et al., Environ. Sci. Technol. **46**, 752 (2012).

Solubility of nonpolar gases in perfluorocarbons





Perfluorodecalin

FIGURE 7.23

Just as the fish are alive and well submerged in water saturated with oxygen, this rodent is alive and well submerged in liquid perfluorodecalin saturated with oxygen. The difference is solubility. There is too little O_2 dissolved in the water for the rodent, who would drown if submerged in it. Similarly, there is too much oxygen dissolved in the perfluorodecalin for the fish, who would suffer from oxygen poisoning if they were to swim downward to say hello to the rodent.

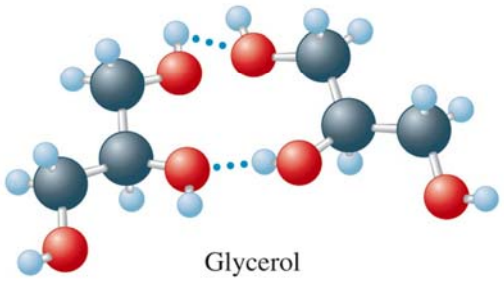
Viscosity

점성도 (viscosity): 흐름에 대한 유체 저항의 척도.

물의 수소결합과 글리세롤의 수소결합은 점성도를 증가시킨다.

TABLE 11.3 Viscosity of Some Common Liquids at 20°C

Liquid	Viscosity ($N\ s/m^2$)*
Acetone (C_3H_6O)	3.16×10^{-4}
Benzene (C_6H_6)	6.25×10^{-4}
Blood	4×10^{-3}
Carbon tetrachloride (CCl_4)	9.69×10^{-4}
Ethanol (C_2H_5OH)	1.20×10^{-3}
Diethyl ether ($C_2H_5OC_2H_5$)	2.33×10^{-4}
Glycerol ($C_3H_8O_3$)	1.49
Mercury (Hg)	1.55×10^{-3}
Water (H_2O)	1.01×10^{-3}



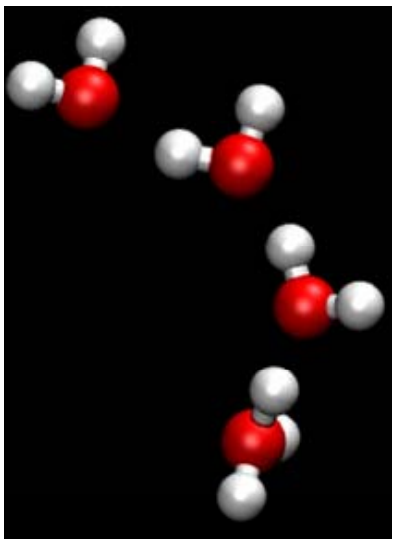
Glycerol





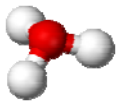
Proton transfer and Grotthuss mechanism

- Grotthuss mechanism

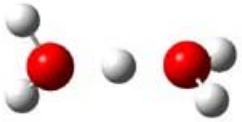


Marx, ChemPhysChem **7**, 1849 (2006)

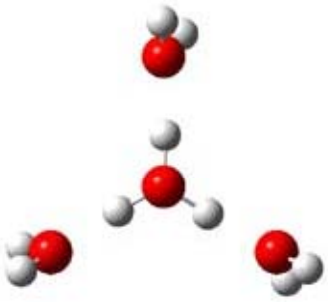
- The structure of hydrated proton
  - Hydronium ion



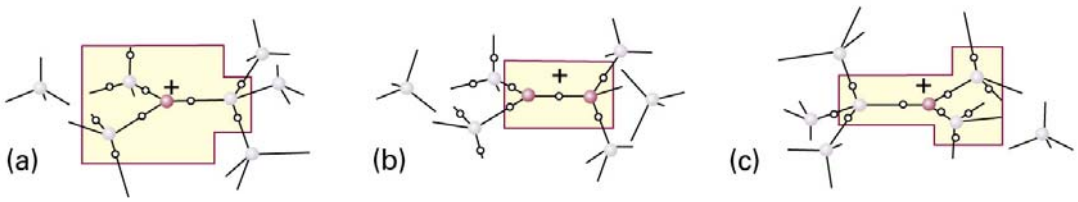
  - Zundel cation ($H_5O_2^+$)



  - Eigen cation ($H_9O_4^+$)



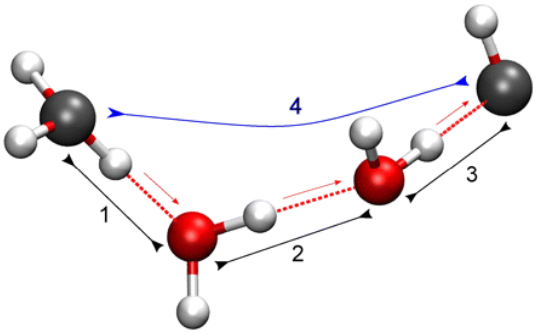
Proposed hopping mechanism in proton transfer



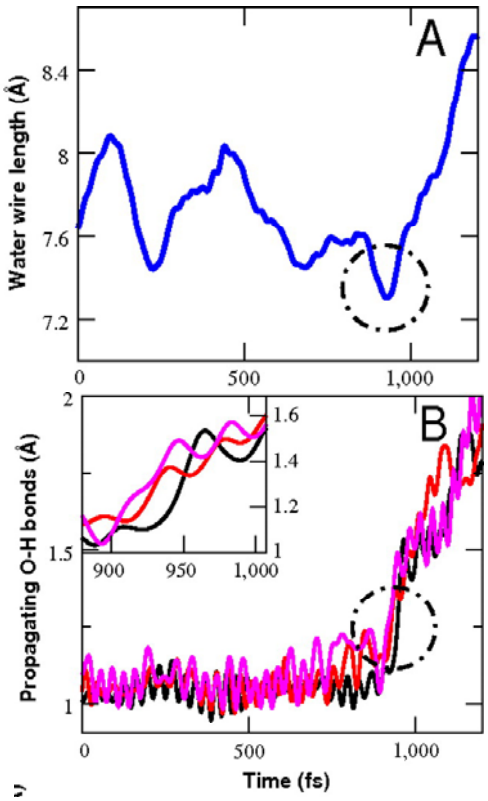
Agmon, Chem. Phys. Lett. **244**, 456 (1995)

Concerted mechanism in the recombination of hydronium and hydroxide ions in water

- Grotthuss mechanism & concerted mechanism in the recommbination reation

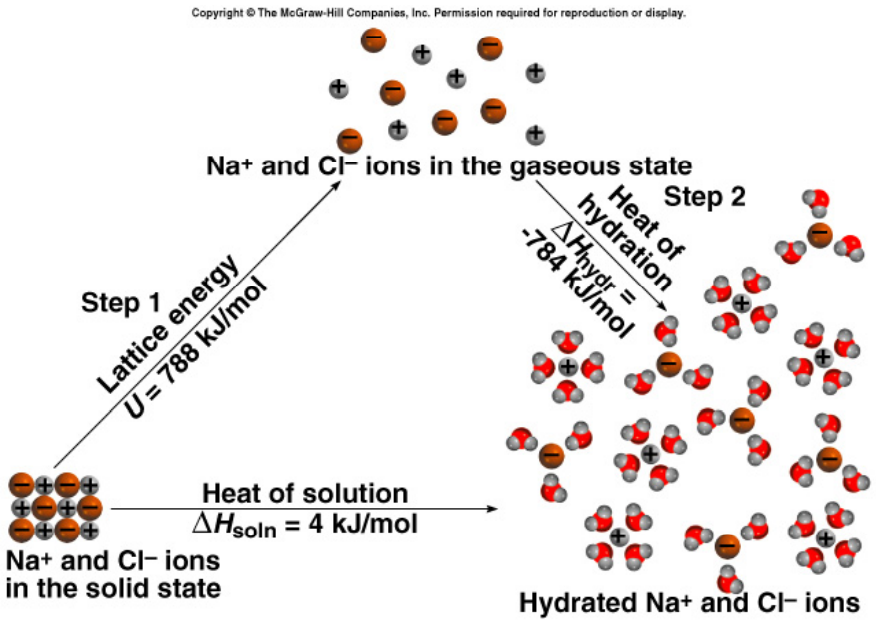


  - time in the formation of water wire: 0.5 ps
  - time in concerted H^+ transfer after the wire formation: 0.065 ps.

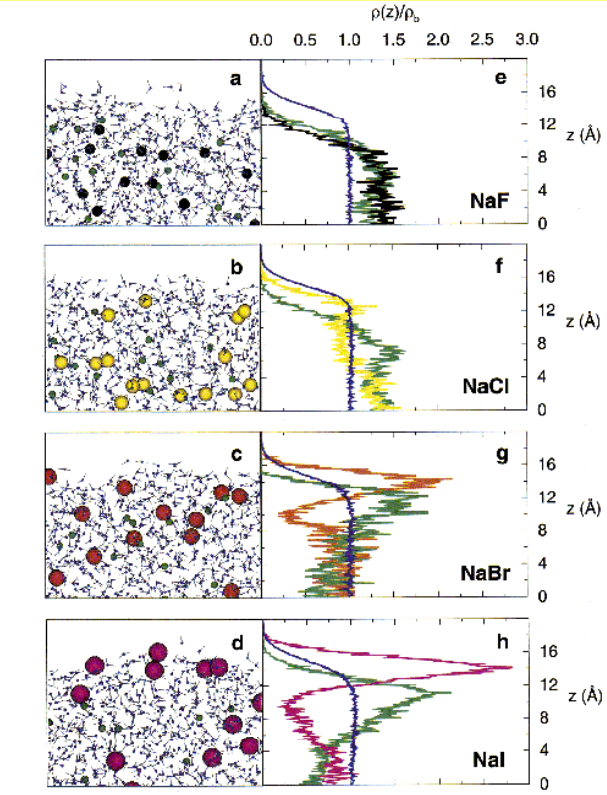


Hassanali et al., Proc. Nat. Aca. Sci. **108**, 20410 (2011)

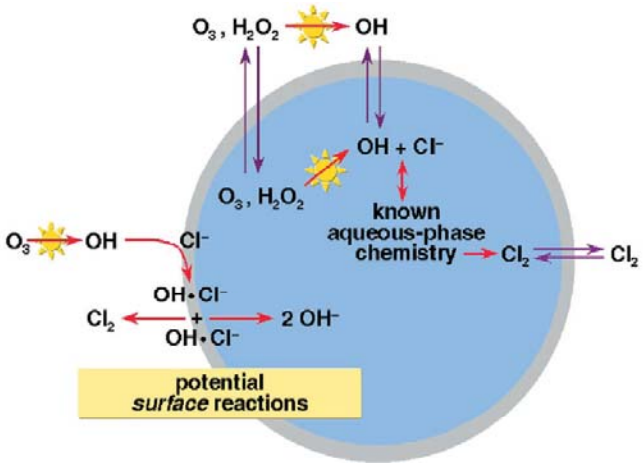
Hydration of ions



Ions at the air / water interface



Jungwirth & Tobias, J. Phys. Chem. B **105,** 10468 (2001)



Knipping et al., Science **288,** 301 (2000)



(예제 1) 다음은 전해질 수용액에 대한 분자동역학 모의실험 결과이다. 전해질 수용액의 표면장력이 물의 표면장력보다 큰 이유는 무엇인가?

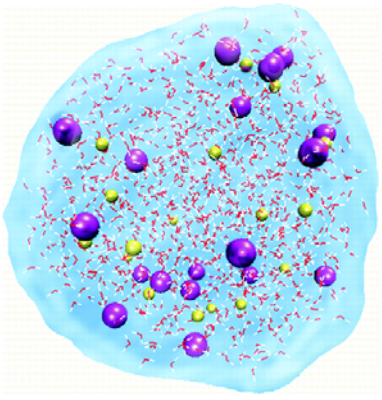


Table 2. Surface Tension of Water Solutions with Sodium Halides[a]

solution	γ (mJ/m^2)	$\Delta\gamma$ (mJ/m^2) simulation	$\Delta\gamma$ (mJ/m^2) experiment[45,46]
NaF	67.5	3.1	3.6
NaCl	67.0	2.6	2.0
NaBr	66.4	2.0	1.6
NaI	65.9	1.5	1.2
H_2O	64.4		

Sun et al., J. Phys. Chem. B **116,** 3198 (2012).

Ionic conductivity

Table 17.1 Limiting ionic conductivities, λ, in water / 10^{-3} S m^2 mol^{-1}

Cations		Anions	
H^+ (H_3O^+)	35.0	OH^-	19.9
Li^+	3.9	F^-	5.5
Na^+	5.0	Cl^-	7.6
K^+	7.4	Br^-	7.8
Cs^+	7.7	I^-	7.7
Ag^+	6.2	HCO_2^-	5.5
NH_4^+	7.4	NO_3^{2-}	7.2
Mg^{2+}	10.6	SO_4^{2-}	13.9
Ca^{2+}	11.9	CO_3^{2-}	17.0
Sr^{2+}	11.9	$CH_3CO_2^-$	4.1

Diffusion Coefficients

- Diffusion: the event where two or more gas or liquid molecules come into contact and mix.

Synoptic table 21.8* Diffusion coefficients at 298 K

	$D/(10^{-9}\ m^2\ s^{-1})$
H^+ in water	9.31
I_2 in hexane	4.05
Na^+ in water	1.33
Sucrose in water	0.522

* More values are given in the *Data section*.

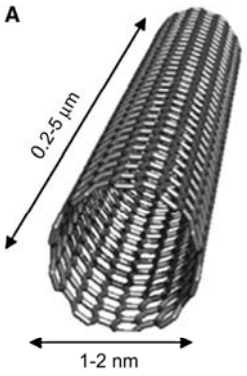
- Relation between ionic conductivity (λ) and diffusion coefficient (D)

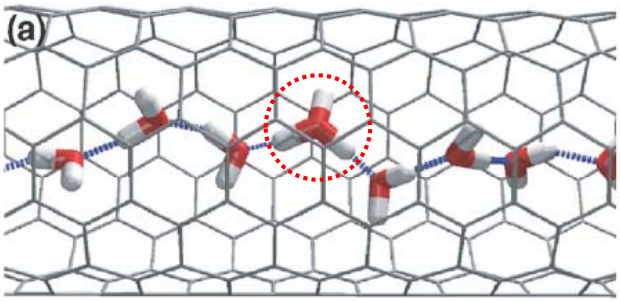
$$\lambda = \frac{z^2 D F^2}{RT}$$

Aktins' Physical Chemistry, Chapt. 21

Proton diffusion in carbon nanotubes

- Carbon nanotube



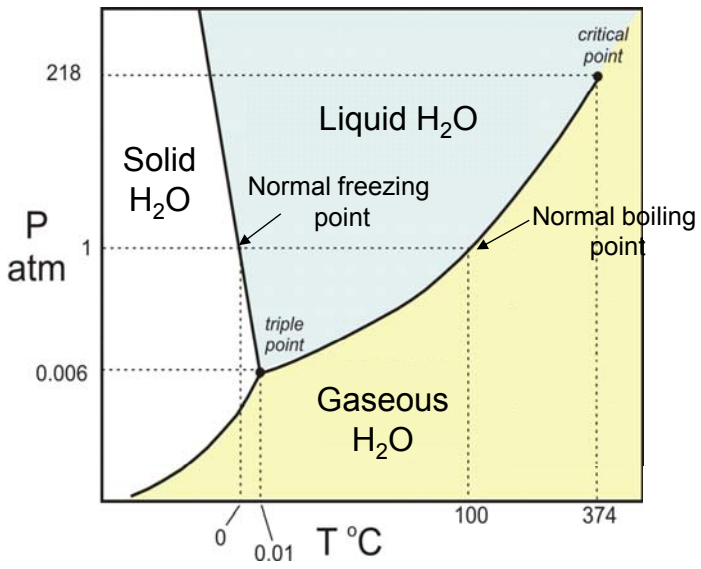


system	D_{H+} (Å^2/ps)
Bulk water (simulation)	0.4
Water in carbon nanotube (simulation)	3 - 17
Bulk water (experiment)	0.93

Dellago et al., Phys. Rev. Lett. **90**, 105902 (2003)

상평형도 (Phase diagram)

- Water phase diagram



Chang, General Chemistry 8th ed., Chapter 11

- 삼중점 (triple point): 세 상태가 서로 평형을 유지하는 유일한 상태.
- 임계온도 (critical temperature, T_c): 이 온도 이상에서는 아무리 높은 압력을 가하더라도 기체가 액화될 수 없는 온도.
- 임계압력 (critical pressure, P_c): 임계온도에서 액화시키기 위해 가해 주는 최소 압력
- 임계점 (critical point): 상평형도에서 임계온도와 임계압력에 의해 정의되는 상태.
- 초임계유체 (supercritical fluid): 임계점을 지나 존재하는 물질의 상태.

Clapeyron and Clasius-Clapeyron Equations

- Clapeyron equation

$$\frac{dP}{dT} = \frac{\Delta_{fus} H}{T \Delta_{fus} V}$$

$$P \approx P^* + \frac{\Delta_{fus} H}{T^* \Delta_{fus} V}(T - T^*)$$

- Clausius-Clapeyron equation

$$\frac{dP}{dT} = \frac{\Delta_{vap} H}{T \Delta_{vap} V}$$

$$\ln\left(\frac{P}{P^*}\right) \approx -\frac{\Delta_{fus} H}{R}\left(\frac{1}{T} - \frac{1}{T^*}\right)$$